Sensitivity Analysis of Class Imbalance and Domain Shift in Vehicle-Dynamics-Based Drowsy Driving Detection

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Abstract—Vehicle-dynamics-based drowsy driving detection (DDD) is a non-intrusive and practical approach to driver monitoring, but its performance may be degraded by two issues: class imbalance, because drowsy samples are rare, and domain shift, because driving behavior varies across drivers. In vehicle-dynamics-based DDD, however, it remains unclear whether these issues materially affect detection performance and which countermeasures are effective. To address this question, we construct a DDD framework using a public 87-driver simulator dataset and conduct a four-factor factorial experiment—varying rebalancing strategy (R), training mode (M), distance metric (D), and domain membership (G)—over 1,512 random forest runs. Sobol–Hoeffding variance decomposition shows that training mode and rebalancing account for more than 95% of systematic variance, with a substantial $R \times M$ interaction that reverses the optimal rebalancing strategy across training modes. In contrast, distance metric (BF_{01} = 71–767) and domain membership are negligible. A subject-wise variant of SMOTE (SW-SMOTE) at ratio $r = 0.1$ with within-domain training achieves the best performance (F2 = 0.558, AUROC = 0.903, AUPRC = 0.648). These results establish a clear design priority for vehicle-dynamics-based DDD: class rebalancing and training mode selection are the dominant performance levers, whereas domain grouping configuration can be safely omitted from the optimization process.

Index Terms—Drowsy driving detection, class imbalance, domain shift, vehicle dynamics, Sobol sensitivity analysis, factorial experiment, SMOTE.

I. Introduction

DROWSY driving is a leading cause of traffic accidents worldwide, accounting for an estimated 15–20% of road fatalities [1]. Vehicle-dynamics-based drowsy driving detection (DDD) offers a non-intrusive and practical approach by monitoring signals such as steering angle, lateral acceleration, and lane offset without requiring sensors attached to the driver [2], [3].

Two challenges are potentially critical for vehicle-dynamics-based DDD. First, class imbalance: drowsy episodes are inherently rare in driving data, with alert states far outnumbering drowsy states, which may bias classifiers toward the majority class and impair detection of safety-critical events. Second, domain shift: driving behavior varies across individuals, and this inter-driver variability may cause performance degradation when a model trained on one group of drivers is applied to another.

Although class imbalance handling [4], [5] and domain adaptation [6], [7] have been studied extensively in other contexts, their importance in vehicle-dynamics-based DDD remains unclear. In particular, it is not known whether these issues materially degrade performance in this modality, whether they warrant dedicated countermeasures, and how they interact when both are present. To this end, we pose the following research questions:

RQ1. To what extent do class rebalancing strategy (R) and training mode (M) affect vehicle-dynamics-based DDD performance?

RQ2. Does the choice of distance metric (D) or domain membership (G) materially impact detection performance?

RQ3. Do the experimental factors interact, and if so, how does the interaction affect practical design decisions?

This paper addresses these research questions by constructing a DDD framework using a public open dataset and conducting a four-factor factorial experiment to quantify the effects of class imbalance handling and domain configuration on detection performance. The four factors—rebalancing strategy (R), training mode (M), distance metric (D), and domain membership (G)—are systematically varied across 1,512 random forest runs, and Sobol–Hoeffding variance decomposition is applied to identify which factors significantly affect performance and which combinations are practically effective. The main contributions are:

1) We construct a DDD experimental framework on a publicly available 87-driver dataset that enables joint, controlled analysis of class imbalance handling and domain shift under a unified factorial design.
2) Through a four-factor factorial experiment with Sobol–Hoeffding variance decomposition, we quantify the relative importance of these design factors and reveal that training mode and class rebalancing dominate ($>$ 95% of systematic variance), with a substantial $R \times M$ interaction that reverses the optimal strategy across training modes.
3) We propose subject-wise SMOTE (SW-SMOTE), which applies oversampling independently within each driver's data to preserve individual driving patterns, and show through exhaustive factor-combination search that SW-SMOTE with within-domain training achieves the best performance (F2

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= 0.558, AUROC = 0.903, AUPRC = 0.648).

The remainder of this paper is organized as follows. Section II reviews related work. Section III formalizes the DDD task and the two data challenges (class imbalance and domain shift). Section IV describes the four-factor experimental design, common settings, and analysis methods. Section V reports factual results from OFAT, Sobol decomposition, interaction analysis, and robustness checks. Section VI answers the research questions and discusses their practical implications. Section VII concludes with a summary and future directions.

II. Related Work

A. Vehicle-Dynamics-Based Drowsy Driving Detection

Vehicle dynamics signals—steering wheel angle, lateral acceleration, and lane position—are widely used for DDD because they enable non-intrusive monitoring. Arefnezhad et al. [8] used statistical and spectral features with SVM, Wang et al. [9] used LSTM with five vehicle signals, and Zhao et al. [10] used continuous wavelet transform to capture transient driving changes. More recently, Zhu et al. [11] proposed MADFM, a multi-layer adaptive fatigue monitoring model based on steering wheel signals. Recent surveys [2], [3], [12], [13] confirm the practical relevance of this modality. However, most studies evaluate a single method on a fixed split, and the sensitivity of detection performance to data preprocessing choices has received little attention.

B. Class Imbalance in Drowsy Driving Detection

Class imbalance is inherent in drowsy driving data because drowsy episodes are rare but safety-critical. He and Garcia [5] summarize major countermeasures such as random under-sampling (RUS), Synthetic Minority Over-sampling Technique (SMOTE) [4], and cost-sensitive learning, but none of these standard techniques account for the multi-subject structure of pooled driving data. In vehicle-dynamics-based DDD, only a few studies explicitly mention imbalance handling: Yadawadkar et al. [14] compared SMOTE with no resampling, Schwarz et al. [15] used SMOTE in steering-based detection, and Baccour et al. [16] used class weighting when comparing vehicle-dynamics-based and driver-based features. However, these studies treat imbalance handling as an auxiliary choice rather than the main object of study, and systematic comparison of multiple countermeasures remains lacking.

C. Domain Shift in Drowsy Driving Detection

Inter-subject variability creates domain shift in DDD [6], [17]. In vehicle-dynamics-based DDD and related driver-state studies, Sun et al. [18] examined subject-independent drowsiness detection while highlighting driver fingerprinting differences, and Maritsch et al. [19] used leave-one-subject-out evaluation for vehicle-dynamics-based driver-state prediction. These studies recognize cross-subject variability, but systematic comparison of domain-related design choices in vehicle-dynamics-based DDD remains limited.

TABLE I
Vehicle Dynamics Signals

Symbol	Signal	Unit
$\delta(t)$	Steering angle	rad
$\dot{\delta}(t)$	Steering speed	rad/s
$a_y(t)$	Lateral acceleration	m/s^2
$a_x(t)$	Longitudinal acceleration	m/s^2
$e_{\text{lane}}(t)$	Lane offset	m

D. Sensitivity Analysis in Machine Learning

Variance-based sensitivity analysis using Sobol indices [20] decomposes output variance into additive contributions of input factors and their interactions. It has been widely applied in engineering and biological modeling [21]. In machine learning, Taylor et al. [22] used Sobol decomposition to show that data-related hyperparameters dominate architectural choices in deep learning pipelines. Probst et al. [23] applied functional ANOVA to quantify the relative importance of random forest hyperparameters, demonstrating that a small subset of tuning decisions accounts for most performance variance. Van Rijn and Hutter [24] performed a large-scale hyperparameter importance study across hundreds of datasets, confirming that the effect of individual hyperparameters varies with dataset characteristics—a finding analogous to our observation that the optimal rebalancing strategy depends on training mode. Despite these advances, Sobol-based sensitivity analysis has not been applied to DDD, particularly to quantify the joint and interactive importance of class imbalance handling and domain configuration.

III. Problem Formulation

This section presents the DDD framework (Section III-A), then formalizes the two data challenges—class imbalance (Section III-B) and domain shift (Section III-C)—that motivate the factorial experiment in Section IV.

A. DDD Framework Overview

The publicly available driving dataset of Aygun et al. [25] is used, comprising 87 subjects each recorded over two sessions on the SIMlsl driving simulator at $f_s = 60$ Hz. Five raw signals are extracted (Table I):

Under the kinematic bicycle model [26], lateral acceleration follows:

$$a_y = \frac{v^2}{L}\,\delta \tag{1}$$

where v is vehicle speed and L is the wheelbase. Lane offset evolves as:

$$\ddot{e}_{\text{lane}}(t) = a_y(t) - a_{y,\text{road}}(t) \tag{2}$$

Because $\dot{\delta} = d\delta/dt$ and $a_y \propto \delta$, four of the five signals (δ, $\dot{\delta}$, a_y, e_{lane}) are deterministically coupled through a single degree of freedom $\delta(t)$, with only a_x dynamically independent.

Raw signals are segmented into 3-second sliding windows with 50% overlap (1.5-second step). Each window is

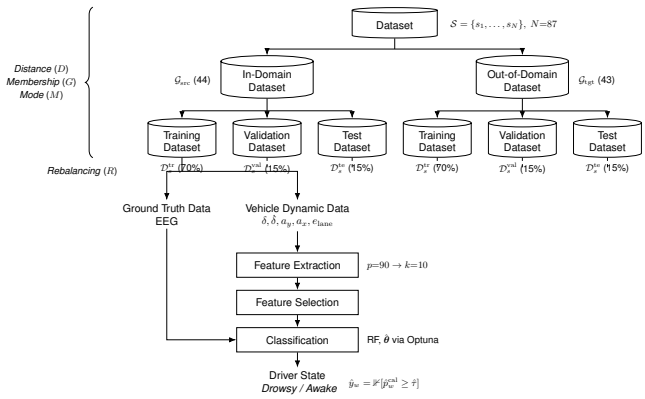

Fig. 1. Experimental framework overview. Vehicle dynamics signals from the 87-subject dataset are segmented into windows and labeled via KSS thresholds. Features are extracted, ranked by RF importance (p=90 → k=10), and classified by an Optuna-tuned random forest. The four experimental factors—distance metric (D), domain membership (G), training mode (M), and rebalancing strategy (R)—determine each factorial cell.

labeled using the Karolinska Sleepiness Scale (KSS) [27]: KSS $\in \{1, \ldots, 5\}$ maps to Alert (class 0) and KSS $\in \{8, 9\}$ to Drowsy (class 1); intermediate scores (6, 7) are excluded to maximize label reliability.

For each window, the 18 statistical and spectral descriptors of Arefnezhad et al. [8] are extracted from each of the five vehicle signals (δ, $\dot{\delta}$, a_y, a_x, e_{lane}), yielding $p = 90$ candidate features. Each signal's raw waveform and its FFT power spectrum in the $[0.5, 30]$ Hz band produce 18 descriptors: Range, StdDev, Energy, Zero-Crossing Rate, Quartiles (Q25, Median, Q75), Skewness, Kurtosis, Katz Fractal Dimension, Shannon Entropy, Frequency Variance, Spectral Entropy, Spectral Flux, Frequency Center of Gravity, Dominant Frequency, Average PSD, and Sample Entropy. To reduce dimensionality and mitigate overfitting, features are ranked by random forest feature importance [28] (200 trees, balanced class weights) and the top $k = 10$ are retained for each training configuration, balancing discriminative power against overfitting given the per-cell sample sizes.

Let $\mathcal{S} = \{s_1, \ldots, s_N\}$ denote the set of $N = 87$ subjects. The windowing, labeling, and feature extraction procedures above yield, for each subject s, a dataset $\mathcal{D}_s = \{(\mathbf{x}_w, y_w)\}_{w=1}^{n_s}$, where $\mathbf{x}_w \in \mathbb{R}^p$ is a feature vector and $y_w \in \{0, 1\}$ is the binary drowsiness label. The DDD task is to learn a classifier $f\colon \mathbb{R}^p \to \{0, 1\}$ from pooled training data that generalizes across subjects. Fig. 1 illustrates the overall processing pipeline.

B. Class Imbalance

Let $n_c = |\{w : y_w = c\}|$ denote the number of windows with label c in a training set. Because drowsy episodes are rare, $n_1 \ll n_0$, yielding an imbalance ratio $\rho = n_1/n_0 \ll 1$. Rebalancing strategies adjust this ratio toward a target value r by under-sampling the majority class or over-sampling the minority class.

TABLE II
Factorial Experiment Design

Factor	Sym.	Lvl.	Values
Rebalancing	R	7	BL, RUS/SMOTE/SW-SMOTE $r \in \{0.1, 0.5\}$
Training mode	M	3	Cross, Within, Mixed
Distance	D	3	MMD, DTW, Wasserstein
Membership	G	2	In-domain, Out-domain

C. Domain Shift

Each subject induces a marginal feature distribution $P_s(\mathbf{x})$. Inter-driver variability means that $P_{s_i}(\mathbf{x}) \neq P_{s_j}(\mathbf{x})$ in general. When subjects are partitioned into a source group $\mathcal{G}_{\text{src}}$ and a target group $\mathcal{G}_{\text{tgt}}$ ($\mathcal{G}_{\text{src}} \cap \mathcal{G}_{\text{tgt}} = \emptyset$), a classifier trained on $\bigcup_{s \in \mathcal{G}_{\text{src}}} \mathcal{D}_s$ may degrade on $\bigcup_{s \in \mathcal{G}_{\text{tgt}}} \mathcal{D}_s$ due to this distributional mismatch.

These two issues are treated here not as fixed nuisances but as explicit design factors whose practical importance is quantified through the factorial experiment in Section IV.

IV. Experimental Framework

This section describes the factorial experiment. Section IV-A provides an overview; Section IV-B defines each factor; Section IV-C describes common experimental settings; and Section IV-D presents the analysis methods.

A. Overview of the Factorial Experiment

The experiment follows a four-factor full-factorial design as shown in Table II.

Each factorial cell is determined by a quadruple (R, M, D, G). The factors form a dependency chain: D determines the domain partition $(\mathcal{G}_{\text{src}}, \mathcal{G}_{\text{tgt}})$ via the KNN median split (Section IV-B1); G selects the evaluation-target group; M assembles the training pool from that partition; and R transforms the pool before classifier fitting. Each subject's dataset $\mathcal{D}_s$ is split chronologically into training, validation, and test portions $(\mathcal{D}_s^{\text{tr}}, \mathcal{D}_s^{\text{val}}, \mathcal{D}_s^{\text{te}})$ with a 70/15/15 ratio (detailed in Section IV-C2).

B. Factor Definitions

1) Distance Metric (D) and Domain Grouping: To instantiate the source/target partition $(\mathcal{G}_{\text{src}}, \mathcal{G}_{\text{tgt}})$ defined in Section III, pairwise distances $d(s_i, s_j)$ between all 87 subjects are computed from the per-subject window sets $\mathcal{D}_{s_i}$ and $\mathcal{D}_{s_j}$ using three metrics:

Maximum Mean Discrepancy (MMD) [29] measures distributional difference in a reproducing kernel Hilbert space $\mathcal{H}$:

$$\text{MMD}^2(s_i, s_j) = \left\| \frac{1}{n_{s_i}} \sum_{w \in s_i} \phi(\mathbf{x}_w) - \frac{1}{n_{s_j}} \sum_{w \in s_j} \phi(\mathbf{x}_w) \right\|_{\mathcal{H}}^2 \tag{3}$$

where ϕ is the feature map induced by a Gaussian kernel.

Dynamic Time Warping (DTW) [30] aligns two univariate time series and finds the warping path that minimizes cumulative Euclidean distance. Each subject's multivariate window sequence is first reduced to a scalar series $\bar{x}_t^{(i)} = \frac{1}{p}\sum_{d=1}^{p} x_{t,d}^{(i)}$ (mean across all z-scored features at each time step), and DTW is computed as:

$$\mathrm{DTW}(s_i, s_j) = \sqrt{\min_{\pi\in\mathcal{P}} \sum_{(a,b)\in\pi} \left(\bar{x}_a^{(i)} - \bar{x}_b^{(j)}\right)^2} \tag{4}$$

where $\mathcal{P}$ is the set of admissible warping paths. Because DTW with Euclidean cost is symmetric, no explicit symmetrization is required.

Wasserstein Distance [31] computes the minimum transport cost between the empirical feature distributions:

$$W_1(s_i, s_j) = \inf_{\gamma\in\Gamma(P_{s_i}, P_{s_j})} \int \|\mathbf{x} - \mathbf{x}'\| \, d\gamma(\mathbf{x}, \mathbf{x}') \tag{5}$$

where $\Gamma(P_{s_i}, P_{s_j})$ is the set of couplings with marginals P_{s_i} and P_{s_j}.

For each metric, a K-nearest-neighbor (KNN) score ($K = 5$) is computed per subject as the mean distance to its K nearest neighbors:

$$\mathrm{KNN}_K(s) = \frac{1}{K} \sum_{j\in\mathcal{N}_K(s)} d(s, s_j) \tag{6}$$

Subjects are split at the median of KNN_K into in-domain (44 subjects, lowest scores, most typical) and out-domain (43 subjects, highest scores, most dissimilar), forming the concrete $\mathcal{G}_{\mathrm{src}}$ and $\mathcal{G}_{\mathrm{tgt}}$ used in the factorial design.

Distance metric $D \in \{\mathrm{MMD}, \mathrm{DTW}, W_1\}$ determines the pairwise subject distances and the binary domain partition $(\mathcal{G}_{\mathrm{src}}, \mathcal{G}_{\mathrm{tgt}})$ via Eqs. (3)–(6).

2) Domain Membership (G): Domain membership $G \in \{\mathrm{in}, \mathrm{out}\}$ selects the evaluation-target group:

$$\mathcal{G}_{\mathrm{eval}} = \begin{cases} \mathcal{G}_{\mathrm{src}} & \text{if } G = \mathrm{in} \text{ (44 subjects)} \\ \mathcal{G}_{\mathrm{tgt}} & \text{if } G = \mathrm{out} \text{ (43 subjects)} \end{cases} \tag{7}$$

3) Training Mode (M): Training mode $M \in \{\mathrm{Cross}, \mathrm{Within}, \mathrm{Mixed}\}$ assembles the training pool from subjects determined by $\mathcal{G}_{\mathrm{eval}}$:

$$\mathcal{D}_{\mathrm{pool}} = \bigcup_{s\,\in\,\mathcal{S}_M} \mathcal{D}_s^{\mathrm{tr}}, \quad \mathcal{S}_M = \begin{cases} \mathcal{S} \setminus \mathcal{G}_{\mathrm{eval}} & M = \mathrm{Cross} \\ \mathcal{G}_{\mathrm{eval}} & M = \mathrm{Within} \\ \mathcal{S} & M = \mathrm{Mixed} \end{cases} \tag{8}$$

Cross-domain trains exclusively on the opposite group, within-domain on the evaluation group itself, and mixed on all 87 subjects.

4) Rebalancing Strategy (R): Rebalancing strategy R applies a rebalancing operator $\mathcal{R}$ to the training pool, yielding $\tilde{\mathcal{D}} = \mathcal{R}(\mathcal{D}_{\mathrm{pool}})$ with adjusted class ratio $\tilde{n}_1/\tilde{n}_0$. Seven levels are defined:

- Baseline (BL): $\tilde{\mathcal{D}} = \mathcal{D}_{\mathrm{pool}}$ (identity; original ratio ρ preserved).
- RUS at $r \in \{0.1, 0.5\}$: randomly discard majority-class samples until $\tilde{n}_1/\tilde{n}_0 = r$.
- SMOTE [4] at $r \in \{0.1, 0.5\}$: generate synthetic minority-class samples via k-NN interpolation on $\mathcal{D}_{\mathrm{pool}}$ until $\tilde{n}_1/\tilde{n}_0 = r$.
- SW-SMOTE at $r \in \{0.1, 0.5\}$: apply SMOTE independently within each subject before pooling:

$$\tilde{\mathcal{D}} = \bigcup_{s\in\mathcal{S}_M} \mathrm{SMOTE}_r(\mathcal{D}_s^{\mathrm{tr}}) \tag{9}$$

so that synthetic samples preserve individual driving patterns rather than interpolating across subjects.

The classifier f is trained on $\tilde{\mathcal{D}}$ and evaluated on $\mathcal{D}_{\mathrm{test}} = \bigcup_{s\in\mathcal{G}_{\mathrm{eval}}} \mathcal{D}_s^{\mathrm{te}}$. The $7 \times 3 \times 3 \times 2 = 126$ factor combinations are each evaluated over 12 fixed random seeds, yielding $N = 1{,}512$ total observations.

C. Common Experimental Settings

1) Classifier and Hyperparameter Optimization: The classifier $f\colon \mathbb{R}^p \to \{0, 1\}$ introduced in Section III is instantiated as a random forest (RF) [28] and trained on the rebalanced pool $\tilde{\mathcal{D}}$ defined above. Its hyperparameter vector $\boldsymbol{\theta}$ is tuned independently per factorial cell via Optuna [32] with 100 Tree-structured Parzen Estimator (TPE) trials and 3-fold stratified cross-validation:

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}\in\Theta} \mathrm{F2}_{\mathrm{CV}}(\boldsymbol{\theta}; \tilde{\mathcal{D}}) \tag{10}$$

where Θ is the search space defined in Table III. After Optuna selection, the classification threshold τ is swept over the set of unique predicted probabilities derived from the precision–recall curve on the validation set, retaining the value that maximizes F2:

$$\hat{\tau} = \arg\max_{\tau\in[0,1]} \mathrm{F2}\Big(\big\{(\hat{p}_w,\, y_w) : (\mathbf{x}_w, y_w) \in \textstyle\bigcup_{s\in\mathcal{G}_{\mathrm{eval}}} \mathcal{D}_s^{\mathrm{val}}\big\},\ \tau\Big) \tag{11}$$

where $\hat{p}_w = \Pr(y{=}1 \mid \mathbf{x}_w; \hat{\boldsymbol{\theta}})$ is the RF posterior probability. The final model is calibrated via Platt scaling [33] (5-fold) on the combined train + validation data, yielding calibrated probabilities $\hat{p}_w^{\mathrm{cal}}$. During HPO, Optuna selects class_weight $\in$ {balanced, balanced_subsample} to guide tree construction. For the final evaluation, a fixed class weighting of {0:1.0, 1:3.0} is applied to all configurations to ensure a uniform minority-recall emphasis across factorial cells, decoupling the class-weight choice from the rebalancing factor R. Each observation in the factorial experiment is therefore:

$$\begin{aligned} Y_{R,M,D,G} &= \mathrm{Metric}\big(\big\{(\hat{y}_w,\, y_w) : (\mathbf{x}_w, y_w) \in \mathcal{D}_{\mathrm{test}}\big\}\big), \\ \hat{y}_w &= \mathbb{1}\big[\hat{p}_w^{\mathrm{cal}} \geq \hat{\tau}\big] \end{aligned} \tag{12}$$

where $\mathcal{D}_{\mathrm{test}} = \bigcup_{s\in\mathcal{G}_{\mathrm{eval}}} \mathcal{D}_s^{\mathrm{te}}$ and Metric $\in \{\mathrm{F2}, \mathrm{AUROC}, \mathrm{AUPRC}\}$.

Computational cost. Each of the 1,512 training runs (126 cells $\times$ 12 seeds) involves 100 Optuna trials with 3-fold CV, totalling $\approx 4.5 \times 10^5$ model fits. Individual HPC jobs required 4 CPU cores, 8–10 GB RAM, and 6–8 hours wall time; the entire experiment consumed $\approx$ 10,000 core-hours on the university cluster.

TABLE III
Random Forest Hyperparameter Search Space (Optuna, 100 TPE Trials)

Parameter	Range	Type
n_estimators	100–1,500	int
max_depth	{5, 10, 20, 30, 50, 100, None}	cat
min_samples_split	10–100	int
min_samples_leaf	5–50	int
max_features	$\{\sqrt{p}, \log_2 p, 0.05, 0.1, 0.2, 0.3, 0.5\}$	cat
min_wt_frac_leaf	0.0–0.1	float
max_samples	{None, 0.5, 0.7, 0.9}	cat
class_weight	{balanced, balanced_subsample}	cat

Inference latency. A trained RF with $\leq 1{,}500$ trees and $k = 10$ input features produces a prediction in < 1 ms on a single CPU core, well within the 3-second window update interval.

2) Evaluation Metrics: The observation $Y_{R,M,D,G}$ in Eq. (12) is computed with three complementary metrics chosen to capture different aspects of imbalanced classification:

F2-score is the threshold-dependent, recall-weighted harmonic mean of precision (P) and recall (R):

$$\mathrm{F2} = \frac{5\,P\,R}{4\,P + R} \tag{13}$$

The $\beta = 2$ weighting penalizes missed drowsy episodes more heavily than false alarms, reflecting the safety-critical nature of DDD. F2 is used as the primary optimization target in Eqs. (10) and (11).

AUROC (area under the ROC curve) summarizes threshold-free discrimination ability over the full operating range. It complements F2 by assessing whether the model ranks drowsy windows above alert ones regardless of threshold choice.

AUPRC (area under the precision–recall curve) [34] provides a threshold-free measure that is more informative than AUROC under severe class imbalance, because it does not credit correct majority-class predictions.

Together, these three metrics form a minimal sufficient set: one threshold-dependent recall-focused metric (F2), one threshold-free global metric (AUROC), and one threshold-free imbalance-aware metric (AUPRC).

The chronological 70/15/15 split introduced in the preceding subsection preserves temporal order within each subject, preventing data leakage from future windows into the training or validation sets.

D. Analysis Methods

Three analysis methods are applied to the 1,512 evaluation results: an intuitive OFAT scan, a Sobol–Hoeffding variance decomposition, and non-parametric statistical tests.

1) One-Factor-at-a-Time (OFAT) Analysis: Before the quantitative decomposition, we perform an OFAT analysis to build intuition. For each target factor, all other factors are fixed at every possible combination, and the mean performance is plotted across the target factor's levels. Each resulting line represents one fixed condition (averaged over 12 seeds), producing a "spaghetti plot" that reveals: (i) whether the factor's effect is consistent across conditions, (ii) how much the estimate varies depending on which conditions are fixed, and (iii) the magnitude of the factor's effect relative to the spread across conditions.

2) Variance-Based Sensitivity Analysis (Sobol Indices): To quantify the relative importance of each factor and their interactions, we adopt the Sobol–Hoeffding functional ANOVA decomposition [20], [35]. Let the response surface $Y = g(X_1, \ldots, X_k)$ map factor configurations to performance, where $Y \equiv Y_{R,M,D,G}$ (Eq. (12)) and $X_i \in \{R, M, D, G\}$. The total variance admits a unique orthogonal decomposition:

$$V(Y) = \sum_{i=1}^{k} V_i + \sum_{i<j} V_{ij} + \cdots + V_{1,2,\ldots,k} \tag{14}$$

where $V_i = \mathrm{Var}_{X_i}\big[\mathbb{E}_{X_{\sim i}}(Y \mid X_i)\big]$ is the variance of the conditional expectation over factor i alone, and V_{ij} captures the joint effect of (X_i, X_j) not explained by their individual contributions. The first-order Sobol index normalises each contribution:

$$S_i = \frac{V_i}{V(Y)} \tag{15}$$

and the total-order index aggregates the main effect with all interactions involving factor i:

$$S_{Ti} = 1 - \frac{V_{\sim i}}{V(Y)} \tag{16}$$

The difference $S_{Ti} - S_i$ quantifies the fraction of variance attributable to interactions involving factor i. Because our design is balanced, all sum-of-squares terms are computed exactly via marginal means without Monte Carlo sampling. Confidence intervals (95%) are obtained by percentile bootstrap ($B = 2{,}000$) resampling over the 12 seeds.

3) Statistical Framework: Due to non-normality (Shapiro–Wilk rejects normality in 45–71% of cells), all pairwise and multi-group comparisons use non-parametric tests.

Multi-group comparisons. Kruskal–Wallis H [36] tests for differences across k independent groups (e.g., seven rebalancing strategies pooled over modes). Friedman χ^2_F [37] tests for differences across k matched conditions (e.g., strategy rankings within each mode).

Pairwise comparisons. Mann–Whitney U for unpaired tests and Wilcoxon signed-rank for paired tests, both Bonferroni-corrected ($\alpha' = \alpha/m$, $\alpha = 0.05$). Effect sizes use Cliff's δ [38] (< 0.147 negligible, < 0.33 small, < 0.474 medium, ≥ 0.474 large). Post-hoc clustering uses Nemenyi critical difference tests [39].

Null-claim support. Because frequentist tests can only fail to reject H_0, claims of factor irrelevance (e.g., distance metric equivalence) are supplemented with Bayes factors via BIC approximation [40], interpreted on the Jeffreys scale [41].

Global validation. A permutation test ($B = 10{,}000$) validates the overall rebalancing effect. Seed convergence analysis confirms ranking stability across the 12 random seeds.

V. Experimental Results

A. Intuitive Factor Analysis via OFAT

Before the quantitative variance decomposition, we screen each factor's effect using OFAT spaghetti plots (Fig. 2). In each panel, every thin line represents one fixed combination of the other three factors (averaged over 12 seeds); the bold line shows the grand OFAT mean and the gray band indicates ±1 SD. Columns correspond to the four experimental factors; rows show F2-score, AUROC, and AUPRC.

The two left columns show the dominant factors. Rebalancing (R): SMOTE and SW-SMOTE variants consistently outperform Baseline and RUS, but the inter-condition spread ($\Delta = 0.08$–0.55 for F2) reveals strong mode-dependence—cross-domain conditions cluster near floor performance while within/mixed conditions fan upward, suggesting an $R \times M$ interaction. Mode (M): every condition rises monotonically from Cross to Within/Mixed, with a fan-out at Within/Mixed that mirrors the rebalancing effect.

The two right columns show the negligible factors. Distance (D): the OFAT mean is flat across MMD, DTW, and Wasserstein, and condition lines remain nearly parallel. Membership (G): the mean is near-flat, and individual lines diverge in both directions, averaging to near zero.

B. Quantitative Factor Decomposition via Sobol Indices

The Sobol–Hoeffding decomposition (Fig. 3) quantifies each factor's contribution to total variance. The permutation test confirms a significant global rebalancing effect for all three metrics ($p < 0.001$).

Mode contributes the largest main effect ($S_M = 0.31$–0.50), followed by Rebalancing ($S_R = 0.24$–0.29). Both factors participate in a substantial $R \times M$ interaction (21.2% of F2-score variance, 13.8% of AUROC variance). The total-order indices show that Mode accounts for $S_{TM} = 0.48$–0.66 and Rebalancing for $S_{TR} = 0.40$–0.46 of total variance.

In contrast, Distance ($S_{TD} < 0.015$) and Membership ($S_{TG} < 0.031$) are negligible even when interactions are included. Residual variance (seed-to-seed variation), denoted S_ε, accounts for 7.9%–21.9% of total variance.

Defining the systematic (non-residual) variance fraction, where S_ε is the residual (seed-level) variance share:

$$\frac{S_M + S_R + S_{R\times M}}{1 - S_\varepsilon} = \frac{0.823}{0.843} = 97.5\% \text{ (F2)} \tag{17}$$

where S_M=0.368, S_R=0.243, $S_{R\times M}$=0.212, and S_ε=0.157. Analogous ratios are 95.8% (AUROC) and 96.1% (AUPRC), confirming that M, R, and their interaction account for $> 95\%$ of systematic variance across all three metrics.

TABLE IV
Bayes Factors for Distance Metric Equivalence

Metric	BF_{01}	Interpretation
F2-score	767	Extreme evidence for H_0
AUROC	71	Very strong evidence for H_0
AUPRC	381	Extreme evidence for H_0

$R \times M$ interaction concentration. The interaction concentration ratio $C^{(i)}_{R\times M} = S_{R\times M}/(S_{Ti} - S_i)$ shows that 87–96% of each dominant factor's interaction variance is attributable to the $R \times M$ term alone, meaning that the factorial design's $2^k - k - 1 = 11$ interaction terms effectively collapse into a single dominant pairwise effect.

Distance metric equivalence—Bayesian confirmation. Because frequentist tests can only fail to reject H_0, we supplement with Bayes factors:

All Bayes factors (Table IV) provide "very strong" to "extreme" evidence on the Jeffreys scale [41] that the distance metric has no effect, affirming the Sobol result ($S_{TD} < 0.015$).

C. $R \times M$ Interaction Analysis

The Sobol decomposition identified a single dominant interaction ($R \times M$). This section characterizes its structure using two complementary analyses.

1) Performance by Strategy and Mode: Table V shows mean performance disaggregated by rebalancing strategy and training mode. Within-domain and Mixed produce statistically equivalent results ($\delta < 0.05$), while Cross-domain collapses to near-chance levels ($\delta > 0.83$ vs. Within).

2) Mode-Dependent Strategy Ranking: The $R \times M$ interaction manifests as a mode-dependent strategy ranking effect: the identity of the best strategy differs across modes. Per-mode Friedman tests (all $p < 10^{-5}$) show RUS r=0.1 ranks first in cross-domain, while SW-SMOTE r=0.1 ranks first in within-domain and mixed. Cross-mode Spearman ρ confirms the shift:

$$\rho_{\text{cross,within}} = -0.74 \text{ to } -0.89 \tag{18}$$

while within- vs. mixed yields $\rho \geq +0.99$ ($p < 0.001$). Kendall's W across the three modes is 0.17–0.22 (non-significant), confirming that modes do not agree on strategy rankings. Fig. 4 visualizes this reversal.

D. Robustness Validation

Cross-metric concordance. Kendall's $W = 0.618$ across seven evaluation metrics; AUROC–AUPRC $\rho = 0.857$. Results are consistent across three metrics and two sampling ratios (91% directional agreement).

Post-hoc power. The pooled design ($n = 36$) detects $|\delta| \geq 0.53$, well below all reported large effects ($|\delta| > 0.8$).

Seed convergence. Ranking stability (σ_{rank}) decreases monotonically with the number of seeds, reaching 0 (F2, AUPRC) or 0.147 (AUROC) by $k = 11$ of 12 seeds (Fig. 5), confirming sufficient seed count.

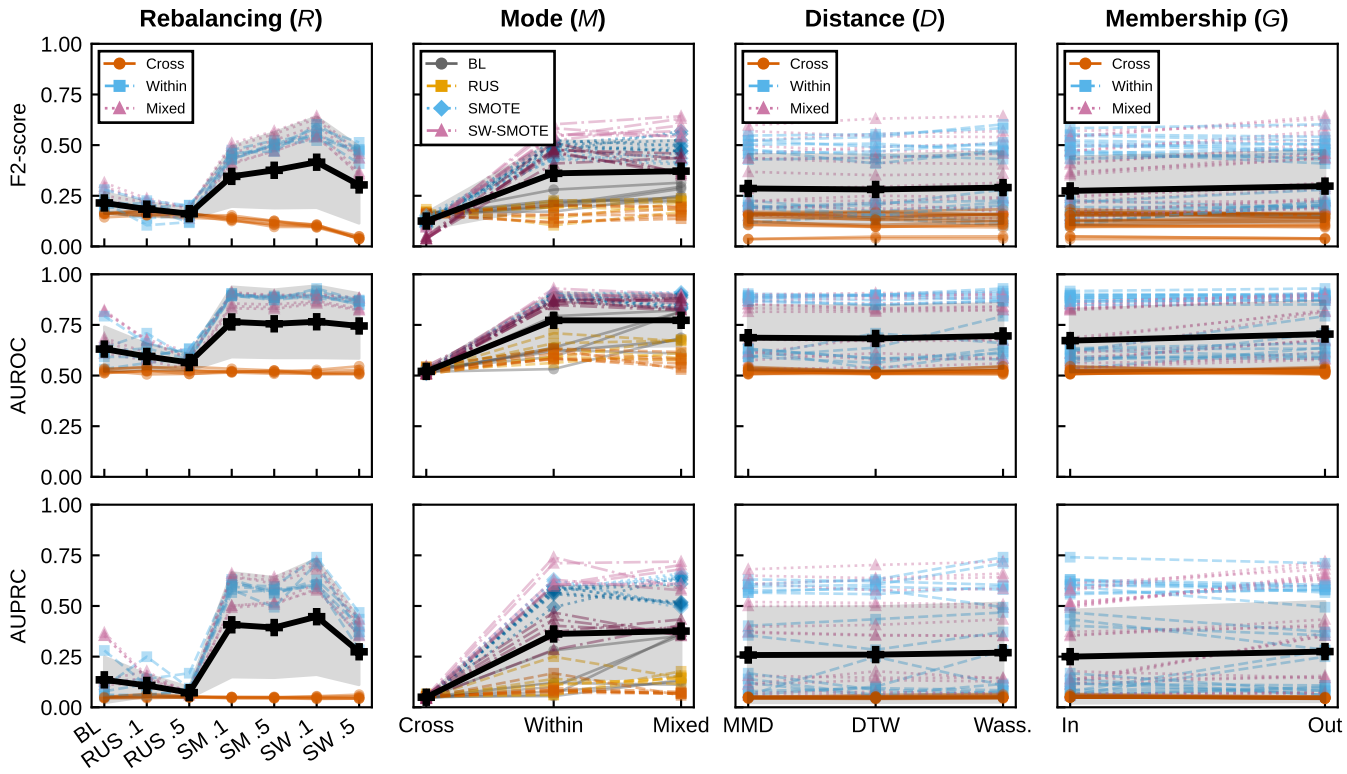

Fig. 2. OFAT spaghetti plots for all four experimental factors (columns: R, M, D, G) and three metrics (rows: F2-score, AUROC, AUPRC). Each thin line represents one fixed combination of the other three factors averaged over 12 seeds. Columns R, D, G encode lines by mode: Cross (vermilion, solid line, circle marker), Within (sky blue, dashed line, square marker), Mixed (reddish purple, dotted line, triangle marker). Column M encodes lines by rebalancing family: BL (grey, solid, circle), RUS (orange, dashed, square), SMOTE (sky blue, dotted, diamond), SW-SMOTE (purple, dash-dot, triangle); ratio variants share the same line style within each family. The bold black line (plus marker) shows the OFAT grand mean; the grey band indicates ±1 SD. Left two columns show dominant factors with large spread; right two columns show flat trajectories confirming negligible effects.

TABLE V
Mean Performance by Rebalancing Strategy, Disaggregated by Training Mode. Cross-domain results (all F2 < 0.17) are shown for completeness; the $R \times M$ interaction analysis focuses on Within and Mixed modes where strategy differentiation is meaningful.

		F2-score		AUROC		AUPRC	
Strategy	F2 (Cross)	Within	Mixed	Within	Mixed	Within	Mixed
Baseline	0.160	0.215	0.268	0.633	0.660	0.116	0.139
RUS $r=0.1$	0.165	0.178	0.208	0.625	0.643	0.122	0.126
RUS $r=0.5$	0.156	0.159	0.170	0.603	0.614	0.097	0.101
SMOTE $r=0.1$	0.138	0.448	0.452	0.898	0.895	0.600	0.593
SMOTE $r=0.5$	0.114	0.499	0.514	0.882	0.879	0.562	0.558
SW-SMOTE $r=0.1$	0.101	**0.558**	**0.587**	**0.903**	**0.901**	**0.648**	**0.641**
SW-SMOTE $r=0.5$	0.042	0.468	0.402	0.862	0.855	0.386	0.371

VI. Discussion

A. Answer to RQ1: Dominance of Training Mode and Rebalancing

The Sobol decomposition reveals a concentrated variance structure: M, R, and $R \times M$ jointly explain $> 95\%$ of systematic variance (Eq. 17), while D and G together contribute $< 5\%$. This concentration can be explained by a mode-dependent performance bottleneck, formalized as follows.

Define the subject overlap ratio between the training pool $\mathcal{S}_M$ (Eq. (8)) and the evaluation group $\mathcal{G}_{\text{eval}}$ (Eq. (7)):

$$\omega_M = \frac{|\mathcal{S}_M \cap \mathcal{G}_{\text{eval}}|}{|\mathcal{G}_{\text{eval}}|} \tag{19}$$

By construction, $\omega_{\text{Cross}} = 0$ (training excludes evaluation subjects), while $\omega_{\text{Within}} = \omega_{\text{Mixed}} = 1$ (evaluation subjects are included). The mode-dependent rebalancing gain, averaged over D and G, is:

$$\Delta Y_R^{(M)} = \mathbb{E}_{D,G}[Y_{R,M,D,G} - Y_{\text{BL},M,D,G}] \tag{20}$$

For the best strategy (SW-SMOTE r=0.1), the empirical

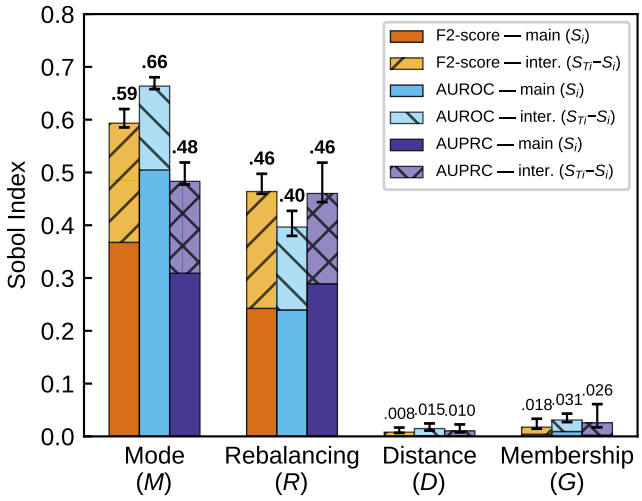

Fig. 3. Sobol sensitivity indices for the four factors. Solid bars: first-order indices (S_i); hatched bars: interaction contributions ($S_{Ti} - S_i$). Error bars: 95% bootstrap CIs on S_{Ti}. Mode and Rebalancing dominate; Distance and Membership are negligible.

values are:

$$\Delta Y^{(M)}_{\text{SW-SM.}} = \begin{cases} +0.343\ (+159\%) & M = \text{Within}\ (\omega = 1) \\ +0.319\ (+119\%) & M = \text{Mixed}\ \ (\omega = 1) \\ -0.059\ (-37\%) & M = \text{Cross}\ \ \ (\omega = 0) \end{cases} \tag{21}$$

When $\omega_M = 1$, the training distribution includes target-subject patterns, so the performance bottleneck is class imbalance ($\rho \ll 1$). SMOTE generates synthetic minority-class samples along the feature manifold, enriching the decision boundary without discarding majority-class information. RUS, by contrast, discards majority-class samples ($\delta_{\text{RUS vs. BL}} = -0.84$), destroying subject-specific calibration.

When $\omega_M = 0$, target-subject data is entirely absent, and the bottleneck shifts to distributional mismatch ($P_{\text{pool}}(\mathbf{x}) \neq P_{\text{test}}(\mathbf{x})$). Oversampling amplifies source-domain patterns that do not transfer, while RUS's information loss is less damaging because the majority-class signal was already misaligned. All strategies collapse to near-chance levels (F2 = 0.04–0.17).

This sign reversal of $\Delta Y^{(M)}_R$ between $\omega = 0$ and $\omega = 1$ (Eq. 21) is the mechanistic origin of the $R \times M$ interaction and the ranking shift (Eq. 18). For comparison, switching distance metric yields $|\Delta\text{F2}| < 0.01$—two orders of magnitude below the rebalancing effect, consistent with $S_{TR}/S_{TD} > 27\times$.

B. Answer to RQ2: Irrelevance of Distance Metric and Domain Membership

Three fundamentally different metrics—MMD (distributional), DTW (temporal), and Wasserstein (geometric)—produce equivalent downstream performance ($S_{TD} < 0.015$, $BF_{01} = 71$–767) despite generating different subject rankings (Table VI; DTW vs. MMD: $\rho = 0.12$, $p = 0.29$).

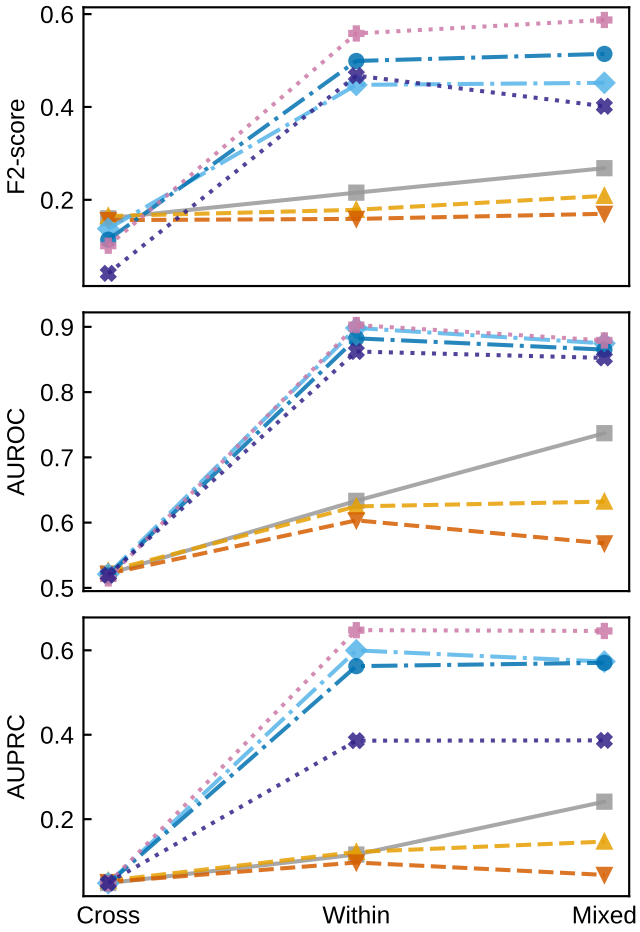

Fig. 4. Mode-dependent strategy performance across training modes. Line styles encode rebalancing families: solid = Baseline (grey), dashed = RUS (orange/vermilion for r=0.1/0.5), dash-dotted = SMOTE (light blue/blue for r=0.1/0.5), dotted = SW-SMOTE (pink/purple for r=0.1/0.5). Markers further distinguish strategies: square, up/down triangle, diamond, circle, plus, and cross. In Cross-domain, all strategies perform poorly; in Within/Mixed, SMOTE-based strategies sharply improve while RUS variants remain low—illustrating the $R \times M$ interaction.

TABLE VI
Pairwise Spearman Correlations Between Distance Metrics

Pair	Spearman ρ
MMD vs. Wasserstein	0.757
MMD vs. DTW	0.116 (n.s.)
Wasserstein vs. DTW	0.444

A three-layer mechanism explains this decoupling. First, the bicycle model (Eqs. 1–2) couples four of the five signals through a single degree of freedom $\delta(t)$, compressing the effective intrinsic dimensionality of the full vehicle-dynamics feature space to $p_{\text{eff}} = 45$ out of 135 total features (95% PCA variance threshold). In this low-dimensional subspace, the three distance functionals converge because the dimensions along which they would disagree carry negligible variance. Second, the intraclass correlation is low (ICC = 0.111), meaning 88.9% of feature variance is intra-subject; consequently, even when

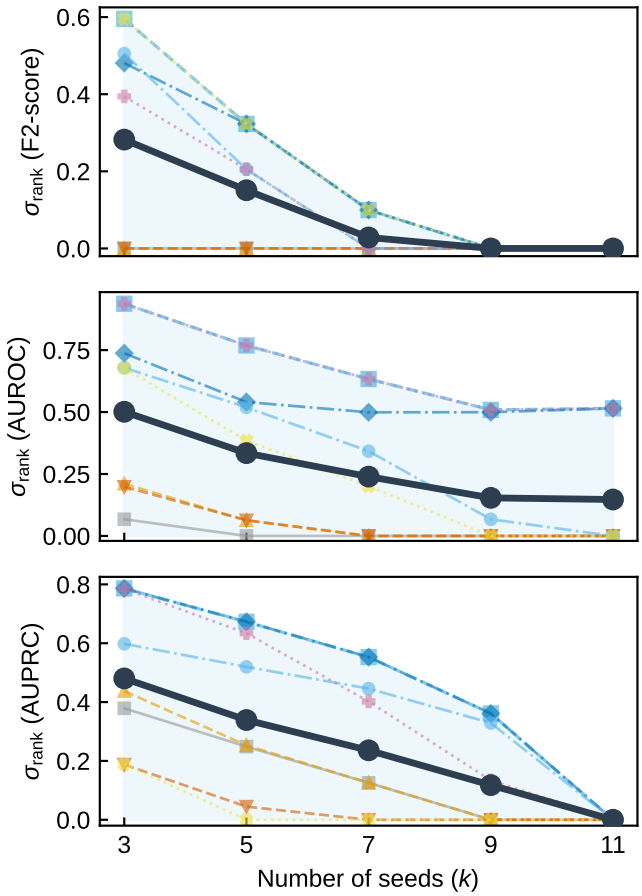

Fig. 5. Ranking stability (σ_{rank}) as a function of seed subset size k. Thick solid line with circles: mean σ_{rank} across all seven strategies; dashed line with squares and shaded band: maximum σ_{rank}. Thin lines show per-strategy convergence (line style encodes family): solid with square = Baseline; dashed with up/down triangles = RUS r=0.1/0.5; dash-dotted with diamond/circle = SMOTE r=0.1/0.5; dotted with plus/cross = SW-SMOTE r=0.1/0.5. Monotonic convergence confirms that $n = 12$ seeds is sufficient.

metric choice relabels up to 42.5% of subjects across domain partitions, the resulting training-set composition changes only marginally. Third, the rebalancing factor dominates residual grouping differences: $S_{TR}/S_{TD} = 27$–$31\times$ (Eq. 22), so once a strategy is chosen, grouping effects are absorbed ($BF_{01} > 70$).

$$\frac{S_{TR}}{S_{TD}} \approx \frac{0.40\text{–}0.46}{< 0.015} = 27\text{–}31\times \tag{22}$$

C. Answer to RQ3: Mode-Dependent Strategy Reversal

The $R \times M$ interaction ($S_{R\times M} = 0.212$ for F2, concentrated at 87–96% of each dominant factor's interaction variance) confirms that the four factors do not act independently. This interaction manifests as a mode-dependent strategy reversal: the identity of the best rebalancing strategy changes across training modes (Eq. 18; Table V). A practitioner who selects a rebalancing strategy based on cross-domain results would deploy the worst strategy for within-domain operation, and vice versa. The mechanistic origin is the sign reversal of $\Delta Y_R^{(M)}$ between $\omega_M = 0$ and $\omega_M = 1$ (Eq. 21): when target-subject data is present ($\omega_M = 1$), the bottleneck is class imbalance and SMOTE-based strategies excel; when absent ($\omega_M = 0$), the bottleneck shifts to distributional mismatch and no strategy helps. Although the domain gap occasionally reverses (out-domain outperforming in-domain), the Sobol decomposition shows this effect is negligible: $S_{G\times M} = 0.006$–0.009, an order of magnitude below $S_{R\times M}$.

D. Practical Implications

The Sobol total-order indices establish a strict factor priority ordering for practitioners deploying vehicle-dynamics-based DDD systems:

$$\begin{aligned} S_{TM}\,(0.48\text{–}0.66) \;&>\; S_{TR}\,(0.40\text{–}0.46) \\ &\gg\; S_{TG}\,(<0.031) \;\approx\; S_{TD}\,(<0.015) \end{aligned} \tag{23}$$

This ordering, combined with the subject overlap ratio ω_M (Eq. 19), yields the following decision rules:

1) Training mode (highest priority): Use within-domain or mixed training whenever target-subject data is available ($\omega_M = 1$).
2) Class rebalancing (second priority): The optimal rebalancing strategy is mode-dependent (Eq. 21). When $\omega_M = 1$, apply SW-SMOTE at $r = 0.1$:

$$R^*(\omega_M) = \begin{cases} \text{SW-SMOTE, } r{=}0.1 & \omega_M{=}1 \\ \text{any (none helps)} & \omega_M{=}0 \end{cases} \tag{24}$$

   where $\Delta Y = +159\%$ (F2) for $\omega_M = 1$ and $\Delta Y \leq 0$ for $\omega_M = 0$.
3) Distance metric and domain membership (negligible): Use any convenient metric for domain grouping. The choice has no measurable impact on classification performance ($BF_{01} > 70$).
4) Bridging the cross-domain gap ($\omega_M = 0$). Since no tested rebalancing strategy improves cross-domain performance (Eq. 21), closing this gap requires reducing the distributional mismatch $P_{\text{pool}}(\mathbf{x}) \neq P_{\text{test}}(\mathbf{x})$ directly. Promising directions include unsupervised domain adaptation [6], [7], few-shot personalization [42], and continual learning. Each can be introduced as an additional factor level within the same Sobol decomposition pipeline, yielding directly comparable variance shares.

E. Comparison with Prior Work

Our finding that class imbalance handling is the dominant design factor aligns with Taylor et al. [22], who showed that data-related hyperparameters dominate architectural choices in deep learning. However, our study is the first to demonstrate this in the DDD context and to reveal the interaction with training mode that causes mode-dependent strategy ranking change. Prior DDD studies that compare rebalancing methods under

TABLE VII
Comparison with Prior DDD Studies. "Factors" Indicates the Number of Design Factors Systematically Varied; "Interaction" Indicates Whether Inter-Factor Interactions Were Quantified. Studies Using the Same 87-Subject Dataset [25] Are Marked with †.

Study	Year	Classif.	Modal.	Subj.	Fact.	Inter.	AUROC
Zhu et al. [11]	2025	Multi-layer	Vehicle	—	0	No	—
Arefnezhad et al. [8]	2019	SVM/NN	Vehicle	20	0	No	0.87
Wang et al. [9]†	2022	LSTM	Vehicle	87	0	No	0.91
Zhao et al. [10]	2012	SVM	Vehicle	15	0	No	0.82
He et al. [43]	2024	GA-CNN	EEG	12	1 (imbalance)	No	0.89
Kim et al. [42]	2025	Mixup	EEG	27	1 (domain)	No	0.92
Ours†	2026	RF	Vehicle	87	4 (R,M,D,G)	$R\times M$ Sobol	0.903

a single training mode [43] risk selecting a suboptimal strategy for deployment.

The best within-domain performance (F2 = 0.558, AUROC = 0.903) is competitive with recent vehicle-dynamics-based methods. Table VII positions this work relative to prior DDD studies. Direct performance comparison is limited because prior studies use different datasets, modalities, and evaluation protocols; the primary contribution here is the factorial decomposition framework, not absolute classification performance. Nevertheless, the within-domain AUROC (0.903) is competitive with Wang et al. [9] (0.91 on the same dataset using an LSTM with a single fixed split), confirming that the RF pipeline optimized via Optuna provides a credible experimental baseline.

The distance metric irrelevance contrasts with the implicit assumption in domain adaptation literature [6], [42] that domain definition substantially impacts downstream performance. As discussed in Section VI, the three-layer mechanism—low effective dimensionality ($p_{\text{eff}} = 45/135$), weak inter-subject discrimination (ICC = 0.111), and rebalancing absorption ($S_{TR}/S_{TD} > 27\times$)—ensures that even substantial partition relabelling produces no measurable change in classification performance.

F. Limitations

1) Single classifier family. The factorial experiment uses RF [28]. The Sobol–Hoeffding decomposition framework itself is classifier-agnostic—it treats the classifier as a black box mapping factor configurations to performance scores—and can therefore be extended to gradient boosting methods, sequence models (LSTM [9]), attention-based architectures [44], or any other learner without methodological modification. Empirical validation with such architectures remains an open question.
2) Deterministic data split. The train/test partition is deterministic; random seeds vary only model initialisation and resampling.
3) Simulator data. All data come from a laboratory driving simulator. The five vehicle-dynamics signals used here (δ, $\dot{\delta}$, a_y, a_x, e_{lane}) are governed by the same bicycle model (Eqs. 1–2) in both simulated and real vehicles; the physical coupling that produces feature redundancy and low ICC (0.111) is intrinsic to vehicle dynamics, not an artifact of the simulator. Nevertheless, on-road conditions introduce additional variability (road surface, weather, traffic interactions) that may alter the absolute variance shares. Validation on naturalistic driving datasets—such as the SHRP 2 corpus—is an essential next step to confirm that the factor hierarchy $M > R \gg D \approx G$ transfers to real-world conditions.
4) Sample size per cell. With $n = 12$, only large effects ($|\delta| > 0.53$) are detectable after Bonferroni correction. Subtle distance metric effects may exist below the detection threshold, though the Bayes factors ($BF_{01} > 70$) argue strongly against this possibility.
5) Feature selection scope. Only the top $k = 10$ features (by RF importance [28]) from the $p = 90$ candidate features are used per configuration. While this aggressive reduction mitigates overfitting for the per-cell sample sizes in this design, a sensitivity analysis over $k \in \{5, 10, 20\}$ and over the full 135-dimensional feature set (adding smooth/prediction-error and wavelet descriptors) would clarify whether the Sobol hierarchy is robust to feature set composition.
6) Binary labeling threshold. KSS ≤ 5 vs. KSS ≥ 8 excludes intermediate drowsiness (KSS 6–7), which may limit applicability to graded drowsiness detection scenarios.

VII. Conclusion

This study quantifies the relative importance of four design factors in vehicle-dynamics-based DDD through a factorial experiment with Sobol–Hoeffding variance decomposition over 1,512 random forest runs from 87 drivers. Training mode ($S_{TM} = 0.48$–0.66) and rebalancing ($S_{TR} = 0.40$–0.46), including their interaction ($S_{R\times M} = 0.21$), account for > 97% of systematic variance, whereas distance metric ($S_{TD} < 0.015$; $BF_{01} = 71$–767) and domain membership ($S_{TG} < 0.031$) are negligible. The $R \times M$ interaction reverses the optimal strategy across modes ($\rho_{\text{cross,within}} = -0.74$ to -0.89), making SW-SMOTE r=0.1 with within-domain training the best overall configuration (F2 = 0.558, AUROC = 0.903). Practitioners should prioritize training mode and class rebalancing (Eq. 23); domain configuration requires no optimization.

Future work includes validation with other classifiers (LSTM, attention), on naturalistic driving data (SHRP 2), and incorporation of domain adaptation and few-shot methods as additional factor levels.

Data and Code Availability

The driving simulator dataset is publicly available at Harvard Dataverse (DOI: 10.7910/DVN/HMZ5RG) [25]. Analysis code and trained models will be released upon publication.

Acknowledgment

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